# NAVEEN R

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## PROFESSIONAL SUMMARY

AI/ML Engineering student with production-grade system deployment experience, currently maintaining a college-wide scheduling platform serving 1000+ active users. Proven expertise in training 80+ ML models with 97.3% accuracy, architecting RESTful APIs handling 500+ concurrent requests, and applying Bayesian inference for healthcare analytics. Seeking software engineering, AI/ML, or data science internships at top-tier technology companies.

## TECHNICAL SKILLS

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| **Languages:** | Python, R, JavaScript, SQL |
| **AI/ML:** | AutoGluon, Scikit-learn, Pandas, NumPy, Bayesian Regression, XGBoost, CatBoost, LightGBM, Neural Networks |
| **Backend:** | FastAPI, REST API Design, Plumber (R), PostgreSQL, Connection Pooling, Query Optimization |
| **Frontend:** | React (Vite), HTML5, CSS3, Shiny (R) |
| **Cloud/DevOps:** | Oracle Cloud Infrastructure, Git, GitHub, Render, Vercel, CI/CD |
| **Core:** | Data Structures, Algorithms, Operating Systems, Optimization Algorithms, Feature Engineering |

## EDUCATION

**B.Tech in Artificial Intelligence & Machine Learning** | K. Ramakrishnan College of Technology
September 2024 – December 2028 | CGPA: 8.3/10 | Relevant Coursework: Data Structures, Algorithms, Machine Learning, Database Systems, Operating Systems, Probability & Statistics

## PROJECTS

### College Timetable Generator

*FastAPI, React (Vite), PostgreSQL, Render, Vercel*

- Architected and deployed production-grade scheduling system currently serving 1000+ students and faculty .s
- Engineered two-phase optimization algorithm (Greedy initialization + Genetic Algorithm) generating conflict-free timetables while optimizing 15+ soft constraints including workload balance and consecutive class avoidance
- Designed intelligent teacher substitution module using multi-factor weighted scoring (subject expertise, current workload, seniority), reducing manual scheduling effort by 85%
- Built RESTful API with FastAPI handling 500+ concurrent requests with sub-200ms P95 latency; implemented connection pooling and query optimization for PostgreSQL

### Customer Churn Prediction

*Python, AutoGluon, Pandas, Scikit-learn, Jupyter*

- Trained and evaluated 80+ ML models (XGBoost, CatBoost, LightGBM, Neural Networks, Random Forest) using AutoGluon AutoML framework, achieving 97.3% ROC-AUC in under 8 minutes on CPU
- Automated end-to-end ML pipeline including data preprocessing, feature engineering, stratified K-fold cross-validation, and hyperparameter optimization
- Extracted actionable business insights by ranking feature importance and churn drivers, enabling data-driven retention strategies
- Demonstrated proficiency in model selection, ensemble stacking, and production-ready ML workflows

### Lung Cancer Risk Analysis

*R, Shiny, Plumber API, Bayesian Regression*

- Applied Bayesian logistic regression to medical datasets, quantifying prediction uncertainty through credible intervals for clinical decision support
- Developed interactive Shiny dashboard enabling healthcare professionals to visualize risk factors, posterior distributions, and probability outcomes
- Exposed predictive models via REST API using Plumber, facilitating integration with electronic health record systems
- Delivered probabilistic insights supporting evidence-based diagnosis and personalized treatment planning

## CERTIFICATIONS

✓Oracle Cloud Infrastructure 2025
✓Prompt Engineering & Programming with OpenAI
✓Programming Foundations with JavaScript, HTML & CSS
✓Certified Generative AI Professional
✓Operating System Fundamentals